Portfolio Optimization Case Study



Course Instructor

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About Joseph

Joseph is passionate about driving better decision-making by creating, curating, and communicating insights from data.

Experience

A data enthusiast with a background in Retail Banking, he is an advocate for making advanced analytics more accessible to a wider audience by integrating them into business intelligence solutions.

Joseph helps organize the Vancouver Power BI & Modern Excel User Group. An active speaker, he has presented sessions on combining data science with business intelligence at conferences in the Pacific Northwest and across North America.

Course Introduction

Portfolio Optimization Case Study

Learning Objectives

01	Create the portfolio from the imported data	02	Visualize the performance over time	03	Calculate a few key metrics, including the daily returns and portfolio Sharpe ratio

Portfolio Optimization Case Study

In this guided case study, we'll go through a real-world scenario, importing financial data to analyze stocks and optimize a portfolio. Each portfolio will contain two years of data for four stocks.



Equal Weighted Portfolio

- Create the Portfolio
- Visualize the Performance
- Calculate Metrics



Optimized Portfolio

- Prepare 10,000 scenarios
- Identify the Optimal Portfolio
- Visualize the scenarios and Optimal Portfolio

Chapter 1
Equal Weighted Portfolio

Equal Weighted Portfolio

The first portfolio will have an equal weighting between the stocks. To build this portfolio, we will:

1. Add 3 columns to stock data

Normalized Return: Return relative to starting value

Allocation: Normalized Return × 0.25

Position Value: Allocation × $10,000

2. Portfolio Visualization

Create 2 line charts:

- Total portfolio value over time
- Individual stock performance

3. Performance Metrics

- Cumulative Return
- Mean Daily Return
- Standard Deviation (Daily)
- Sharpe Ratio
- Annualized Sharpe Ratio

Chapter 2
Optimized Portfolio

Optimized Portfolio

The second portfolio will be optimized with a weighting allocation that provides the best return, adjusted for risk. To build this portfolio, we will:

01

Scenario Preparation

- 10,000 rows for 4 stocks
- Each row is a unique weight combination

02

Generate Portfolio Scenarios

- Loop through all 10,000 rows
- Assign random weight combinations

03

Identify Optimal Portfolio

- Identify the highest Sharpe ratio
- Record optimal stock weights

04

Visualize Results

- Plot 10,000 portfolios on a scatter plot
- Highlight the point with the highest Sharpe ratio

Course Summary